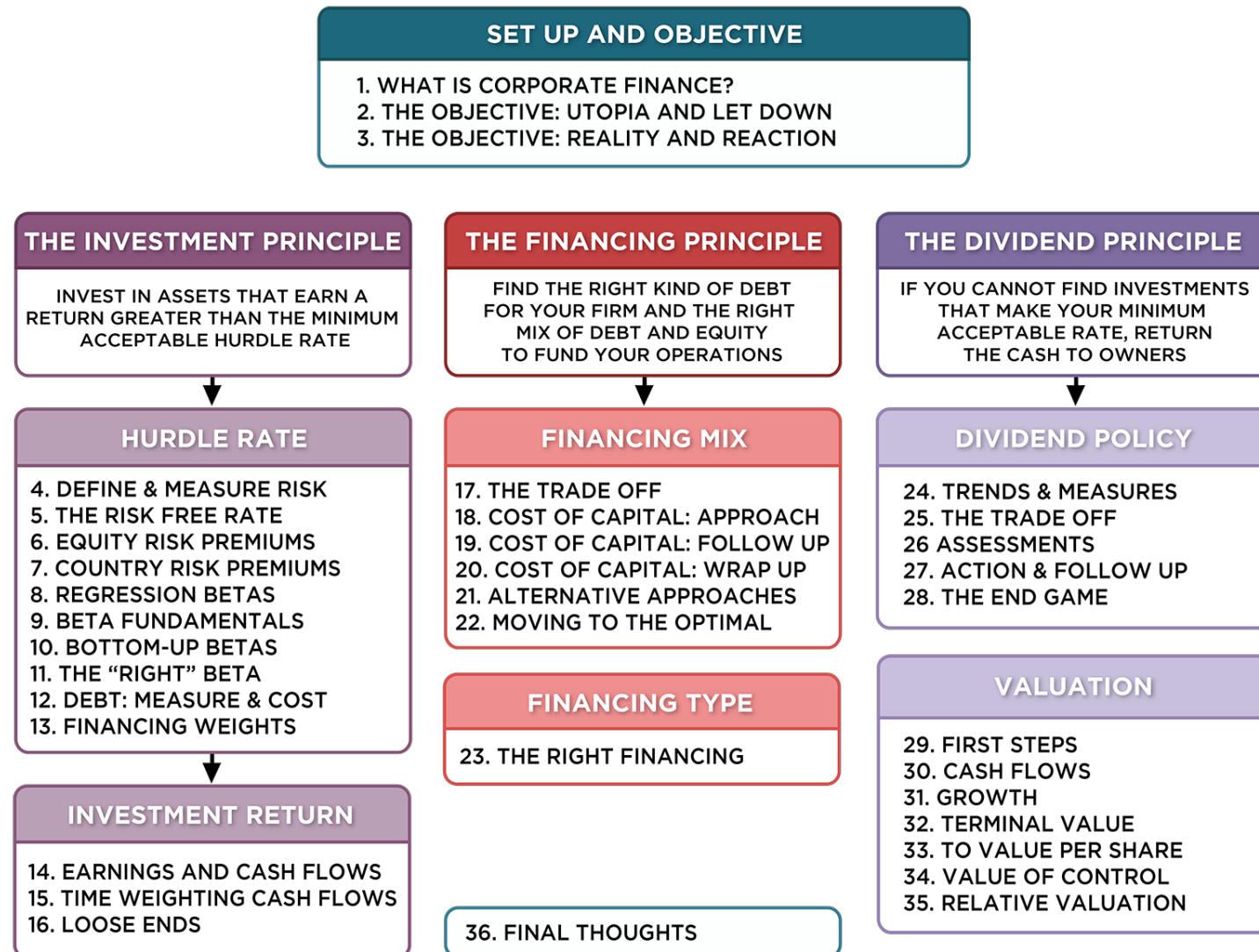


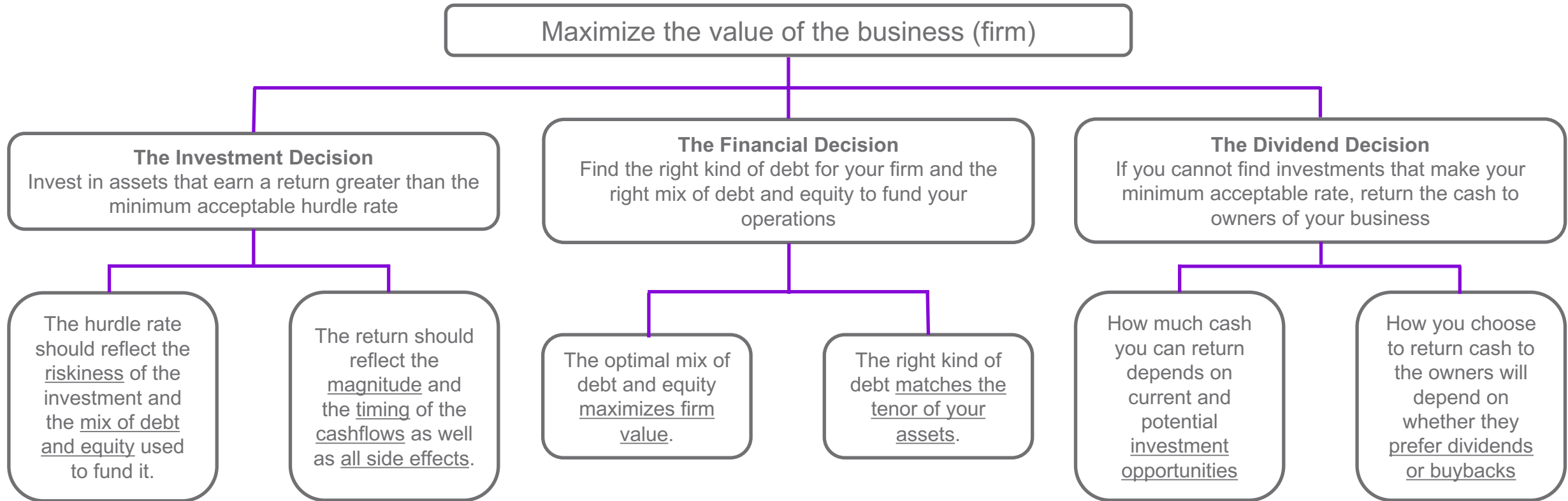
SESSION 04: DEFINE AND MEASURE RISK

Risk = Danger + Opportunity

SESSION 04: DEFINE AND MEASURE RISK



FIRST PRINCIPLES



THE NOTION OF A BENCHMARK

- Since financial resources are finite, there is a hurdle that projects have to cross before being deemed acceptable.
- This hurdle will be higher for riskier projects than for safer projects.

- A simple representation of the hurdle rate is as follows:

$$\text{Hurdle rate} = \text{Riskless Rate} + \text{Risk Premium}$$

- The two basic questions that every risk and return model in finance tries to answer are:
 - How do you measure risk?
 - How do you translate this risk measure into risk premium?

WHAT IS RISK?

- Risk, in traditional terms, is viewed as a ‘negative’. Webster’s dictionary, for instance, defines risk as “exposing to danger or hazard”. The Chinese symbols for risk, reproduced below, give a much better description of risk:

危机

- The first symbol is the symbol for “danger”, while the second is the symbol for “opportunity”, making risk a mix of danger and opportunity. You cannot have one, without the other.
- Risk is therefore neither good nor bad. It is just a fact of life. The question that businesses have to address is therefore not whether to avoid risk but how best to incorporate it into their decision making.

A GOOD RISK AND RETURN MODEL SHOULD . . .

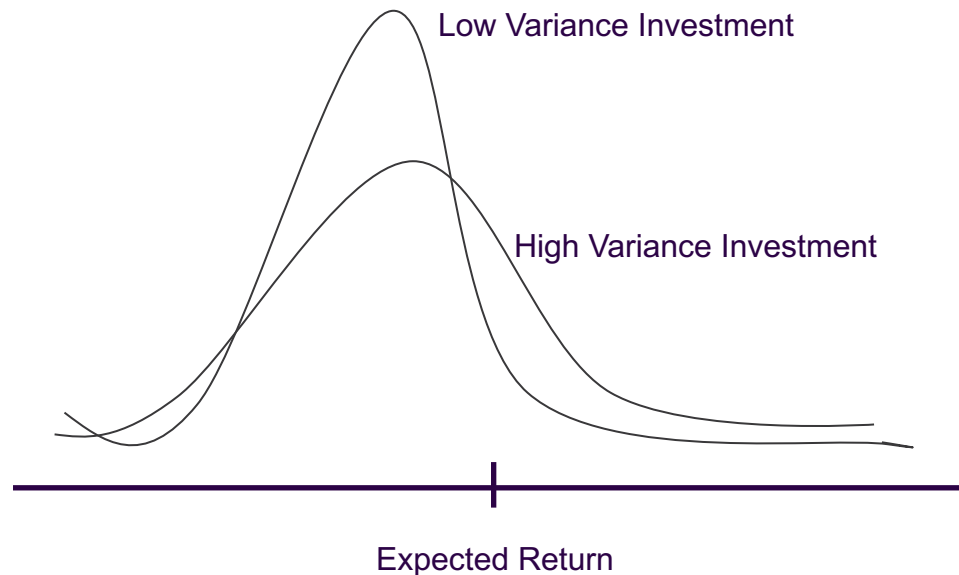
1. It should come up with a measure of risk that applies to all assets and not be asset-specific.
2. It should clearly delineate what types of risk are rewarded and what are not, and provide a rationale for the delineation.
3. It should come up with standardized risk measures, i.e., an investor presented with a risk measure for an individual asset should be able to draw conclusions about whether the asset is above-average or below-average risk.
4. It should translate the measure of risk into a rate of return that the investor should demand as compensation for bearing the risk.
5. It should work well not only at explaining past returns, but also in predicting future expected returns.

THE CAPITAL ASSET PRICING MODEL

1. Uses variance of actual returns around an expected return as a measure of risk.
2. Specifies that a portion of variance can be diversified away, and that is only the non-diversifiable portion that is rewarded.
3. Measures the non-diversifiable risk with beta, which is standardized around one.
4. Translates beta into expected return -
$$\text{Expected Return} = \text{Riskfree rate} + \text{Beta} * \text{Risk Premium}$$
5. Works as well as the next best alternative in most cases.

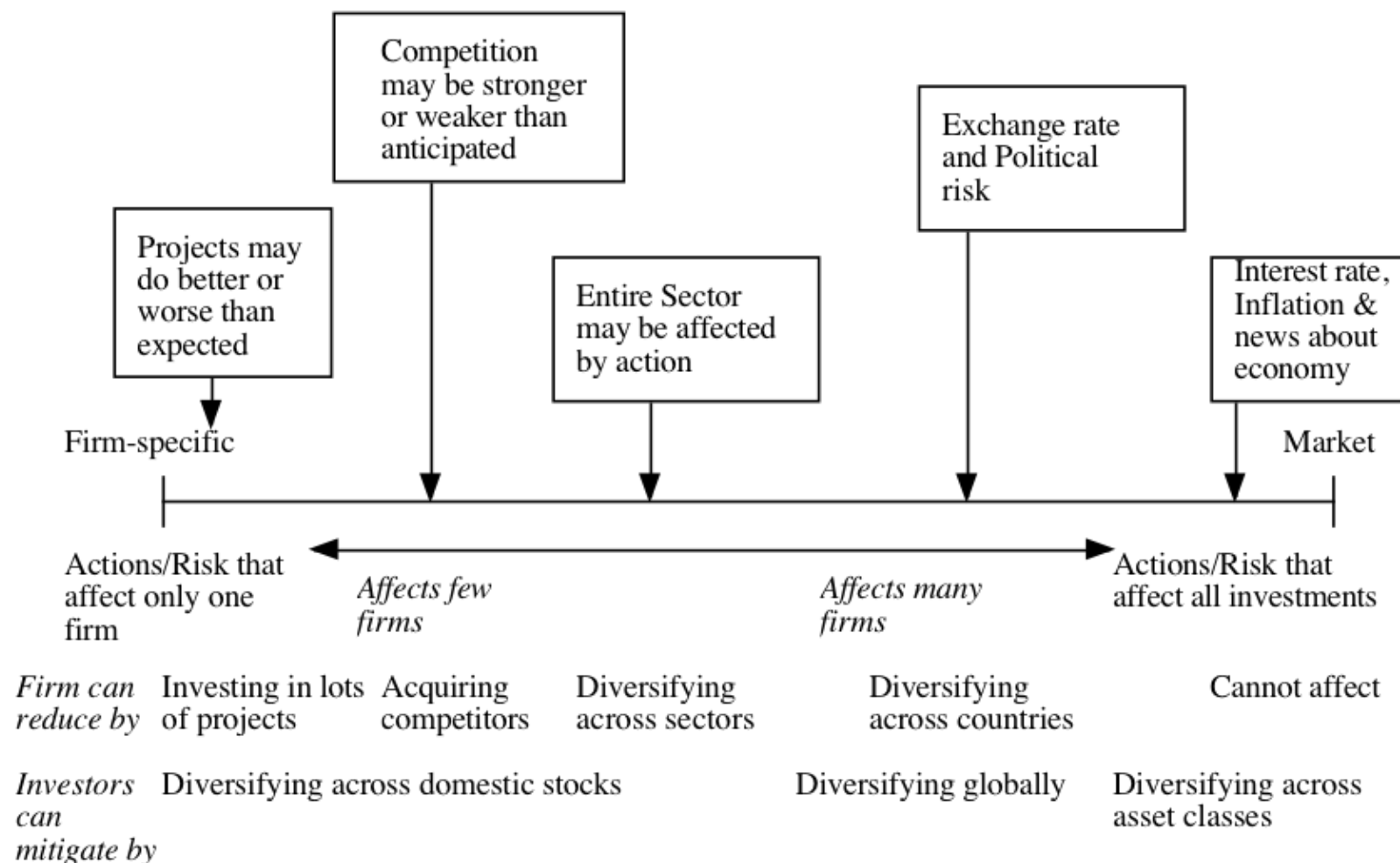
1. THE MEAN-VARIANCE FRAMEWORK

- The variance on any investment measures the disparity between actual and expected returns.



2. THE IMPORTANCE OF DIVERSIFICATION: RISK TYPES

Figure 3.5: A Break Down of Risk



THE ROLE OF THE MARGINAL INVESTOR

- The marginal investor in a firm is the investor who is most likely to be the buyer or seller on the next trade and to influence the stock price.
- Generally speaking, the marginal investor in a stock has to own a lot of stock and also trade that stock on a regular basis.
- Since trading is required, the largest investor may not be the marginal investor, especially if he or she is a founder/manager of the firm (Larry Ellison at Oracle, Mark Zuckerberg at Facebook).
- **In all risk and return models in finance, we assume that the marginal investor is well diversified.**

3. THE LIMITING CASE: THE MARKET PORTFOLIO

- The big assumptions & the follow up: Assuming diversification costs nothing (in terms of transactions costs), and that all assets can be traded, the limit of diversification is to hold a portfolio of every single asset in the economy (in proportion to market value). This portfolio is called the market portfolio.
- The consequence: Individual investors will adjust for risk, by adjusting their allocations to this market portfolio and a riskless asset (such as a T-Bill):

<i>Preferred Risk Level</i>	<i>Allocation Decision</i>
No risk	100% in T-bills
Some risk	50% in T-Bills; 50% in Market Portfolio
A little more risk	25% in T-Bills; 75% in Market Portfolio
Even more risk	100% in Market Portfolio
A risk hog . . .	Borrow money, invest in Market Portfolio

4. THE RISK OF AN INDIVIDUAL ASSET

- The essence: The risk of any asset is the risk that it adds to the market portfolio. Statistically, this risk can be measured by how much an asset moves with the market (called the covariance)
- The measure: Beta is a standardized measure of this covariance, obtained by dividing the covariance of any asset with the market by the variance of the market. It is a measure of the non-diversifiable risk for any asset. Beta can be measured by the covariance of its returns with returns on a market index, which is defined to be the asset's beta.
- The result: The required return on an investment will be a linear function of its beta:
 - Expected Return = Riskfree Rate + Beta * (Expected Return on the Market Portfolio - Riskfree Rate)

ALTERNATIVES TO THE CAPM

Step 1: Defining Risk

The risk in an investment can be measured by the variance in actual returns around an expected return

Riskless Investment



$E(R)$

Low Risk Investment



$E(R)$

High Risk Investment



$E(R)$

Step 2: Differentiating between Rewarded and Unrewarded Risk

Risk that is specific to investment (Firm Specific)

Can be diversified away in a diversified portfolio
1. each investment is a small proportion of portfolio
2. risk averages out across investments in portfolio

Risk that affects all investments (Market Risk)

Cannot be diversified away since most assets are affected by it.

The marginal investor is assumed to hold a “diversified” portfolio. Thus, only market risk will be rewarded and priced.

Step 3: Measuring Market Risk

The CAPM	The APM	Multi-Factor Models	Proxy Models
If there is 1. no private information 2. no transactions cost the optimal diversified portfolio includes every traded asset. Everyone will hold this <u>market portfolio</u> Market Risk = Risk added by any investment to the market portfolio:	If there are no arbitrage opportunities then the market risk of any asset must be captured by betas relative to factors that affect all investments. Market Risk = Risk exposures of any asset to market factors	Since market risk affects most or all investments, it must come from macro economic factors. Market Risk = Risk exposures of any asset to macro economic factors.	In an efficient market, differences in returns across long periods must be due to market risk differences. Looking for variables correlated with returns should then give us proxies for this risk. Market Risk = Captured by the Proxy Variable(s)
Beta of asset relative to Market portfolio (from a regression)	Betas of asset relative to unspecified market factors (from a factor analysis)	Betas of assets relative to specified macro economic factors (from a regression)	Equation relating returns to proxy variables (from a regression)

LIMITATIONS OF THE CAPM

1. The model makes unrealistic assumptions
2. The parameters of the model cannot be estimated precisely
 - Definition of a market index
 - Firm may have changed during the 'estimation' period
3. The model does not work well
 - If the model is right, there should be
 - a linear relationship between returns and betas
 - the only variable that should explain returns is beta.
 - The reality is that
 - the relationship between betas and returns is weak
 - other variables (size, price/book value) seem to explain differences in returns better.

WHY THE CAPM PERSISTS . . .

- The CAPM, notwithstanding its many critics and limitations, has survived as the default model for risk in equity valuation and corporate finance. The alternative models that have been presented as better models (APM, Multifactor model) have made inroads in performance evaluation but not in prospective analysis because:
 - The alternative models (which are richer) do a much better job than the CAPM in explaining past return, but their effectiveness drops off when it comes to estimating expected future returns (because the models tend to shift and change).
 - The alternative models are more complicated and require more information than the CAPM.
 - For most companies, the expected returns you get with the the alternative models is not different enough to be worth the extra trouble of estimating four additional betas.

GAUGING THE MARGINAL INVESTOR: DISNEY IN 2009

DIS US \$ ↑ **24.2422** +.7422 D 2s EquityHDS

DELAY 14:27 Vol 6,135,972 Op 23.81 Z Hi 24.34 T Lo 23.8 T ValTrd 148.014m

DIS US Equity 95) Saved Searches 96) Default Settings Page 1/150 Holdings Search

Walt Disney Co/The CUSIP 25468710

21) Sources 22) Types 23) Countries 24) Metro Areas 25) Advanced Filters

Name Filter Sort By **Mkt Val**

	Holder Name	Portfolio Name	Source	Mkt Val	% Out	Mkt Val Chg	File Dt
1)	JOBS STEVEN PAUL	n/a	Form 4	3.34BLN	7.46	0	5/5/06
2)	FIDELITY MANAGEMENT &	FIDELITY MANAGEMEN	13F	2.05BLN	4.58	-36.12MLN	9/30/08
3)	STATE STREET CORP	STATE STREET CORPO	13F	1.7BLN	3.79	-18.6MLN	9/30/08
4)	BARCLAYS GLOBAL INVES	BARCLAYS GLOBAL IN	13F	1.66BLN	3.70	-160.12MLN	9/30/08
5)	VANGUARD GROUP INC	VANGUARD GROUP IN	13F	1.38BLN	3.08	-6.82MLN	9/30/08
6)	SOUTHEASTERN ASSET M	SOUTHEASTERN ASSE	13F	1.12BLN	2.50	-14.03MLN	9/30/08
7)	STATE FARM MUTUAL AU	STATE FARM MUTUAL	13F	1.02BLN	2.28	0	9/30/08
8)	WELLINGTON MANAGEMEN	WELLINGTON MANAGE	13F	939.38MLN	2.09	110.6MLN	9/30/08
9)	CLEARBRIDGE ADVISORS	CLEARBRIDGE ADVISO	13F	815.91MLN	1.82	-47.04MLN	9/30/08
10)	JP MORGAN CHASE & CO	JP MORGAN CHASE &	13F	693.31MLN	1.55	-18.89MLN	9/30/08
11)	MASSACHUSETTS FINANCI	MASSACHUSETTS FINA	13F	682.16MLN	1.52	112.29MLN	9/30/08
12)	BANK OF NEW YORK MELL	BANK OF NEW YORK	13F	681.68MLN	1.52	-57.13MLN	9/30/08
13)	NORTHERN TRUST CORP	NORTHERN TRUST CO	13F	610.26MLN	1.36	-4.81MLN	9/30/08
14)	AXA	AXA	13F	486.28MLN	1.08	47.05MLN	9/30/08
15)	BLACKROCK INVESTMENT	BLACKROCK INVESTME	13F	476.12MLN	1.06	-47.11MLN	9/30/08
16)	JENNISON ASSOCIATES L	JENNISON ASSOCIATE	13F	428.85MLN	0.96	-102.77MLN	9/30/08
17)	T ROWE PRICE ASSOCIAT	T ROWE PRICE ASSOC	13F	351.61MLN	0.78	-9.94MLN	9/30/08

26) Latest Chg 27) Hist Held % Out on Page **41.12**

Australia 61 2 9777 8600 Brazil 5511 3048 4500 Europe 44 20 7330 7500 Germany 49 69 9204 1210 Hong Kong 852 2977 6000
Japan 81 3 3201 8900 Singapore 65 6212 1000 U.S. 1 212 318 2000 Copyright 2009 Bloomberg Finance L.P.
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EXTENDING THE ASSESSMENT OF THE INVESTOR BASE

- In all five of the publicly traded companies that we are looking at, institutions are big holders of the company's stock.

	Disney	Deutsche Bank	Vale (preferred)	Tata Motors	Baidu (Class A)
Institutions	70.2%	40.9%	71.2%	44%	70%
Individuals	21.3%	58.9%	27.8%	25%	20%
Insiders	7.5%	0.2%	1.0%	31%	10%

	Largest Holder	Number of institutional investors in top 10 holdings
Disney	Laurene Jobs (7.3%)	8
Deutsche Bank	Blackrock (4.69%)	10
Vale (preferred)	Aberdeen (7.40%)	8
Tata Motors	Tata Sons (26.07%)	7
Baidu (Class A)	Capital Group (12.46%)	10

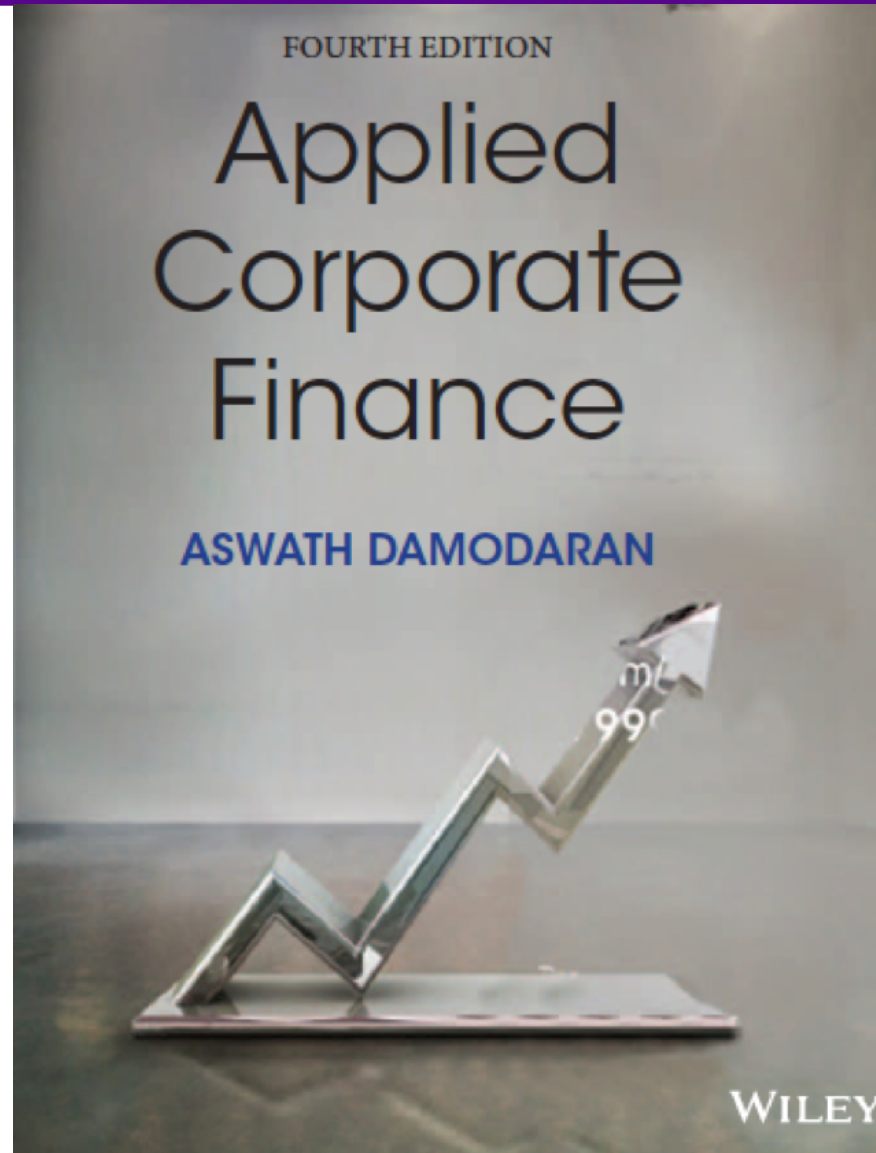


APPLICATION TEST: WHO IS THE MARGINAL INVESTOR IN YOUR FIRM?

- Looking at the breakdown of stockholders in your firm, consider whether the marginal investor is
 - An institutional investor
 - An individual investor
 - An insider

[B DES Page 3](#)
[PB Page 13](#)

Task
Who is the
marginal investor in
your firm?



Optional:
Read
Chapter 3